

Aniket Deshpande

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Areas of Research

Many-Body Physics · Machine Learning

Education

2023–2027

University of Illinois, Urbana-Champaign

B.S. in Physics, *Specialization in Physics & Computation*

Selected Training

Aug.–Dec. 2025

Independent Study, Information Geometry, with Prof. Matthew Singh

Aug. 2025

QSim Summer School — NSF RQS (hosted at IBM, NYC)

May 2025

UQ & ML for Physical Systems — IMSI, University of Chicago

Experience

Fall 2026

Q Labs

San Francisco, CA

Member of Technical Staff

Researching theories of generalization in large language models.

Summer 2026

Lawrence Berkeley National Lab

Berkeley, CA

Student Researcher, *Mentor: Prof. Roel Van Beeumen*

Randomized sketching algorithms for isometric tensor network states.

Feb. 2026–

Supervised Program for Alignment Research

Remote

Research Fellow

Developing temporal crosscoders, a sparse dictionary learning architecture that ties decoder atoms across token positions to exploit multi-position temporal structure in transformer residual stream activations. Evaluating against standard SAEs and layer-wise crosscoders on sparse probing benchmarks and downstream reasoning-steering pipelines.

Feb.–May 2026

Vector Institute for Artificial Intelligence

Toronto, ON

Research Intern, *PI: Prof. Alán Aspuru-Guzik*

Developing geometry-conditioned regressors for fermionic classical shadow data to predict time-resolved molecular observables, with accompanying sample-complexity and time-resolution guarantees.

Sep. 2024–

Lab for Parallel Numerical Algorithms

Urbana, IL

Undergraduate Researcher, *PI: Prof. Edgar Solomonik*

Developing an annealed importance sampling algorithm with Glauber dynamics for approximating tensor network contractions, with provable mixing time guarantees and applications to quantum simulation. Proved that alternating Mahalanobis distance minimization for CP decomposition performs maximum likelihood estimation under a Kronecker-structured noise covariance model.

May–Aug. 2024

Space Dynamics Laboratory

Albuquerque, NM

Ionospheric Analyst Intern

Developed a Python scraper to expedite data collection of NICT ionograms (600+/hour). Implemented and benchmarked noise-reduction filters in Python and Julia (PSNR, MSE, SSIM). Researched CNN architectures for automatic ionogram scaling.

Publications & Preprints

- 2026 **Decoder-Preserving Sparse Autoencoders: Which Readouts Survive Sparse Compression?**
A. Deshpande.
arXiv:2607.17425
- 2026 **Fermionic Shadow Regression: Amortized Learning of Molecular Electron Dynamics**
A. Deshpande, L. Mantilla Calderón, K. Panicker, M. Bagherimehrab, A. Aspuru-Guzik.
4th International Workshop on Quantum Machine Learning at IEEE QCE 2026
- 2026 **Feature-Resolved Attention**
D. Manning-Coe, N. Aljaafari, J. Stephenson, K. Kapre, H. Jain, **A. Deshpande.**
Mechanistic Interpretability at ICML 2026
- 2026 **Crosscoding Through Time: Sparse Feature Discovery Across Sequence Positions**
D. Manning-Coe, H. Xuanyuan, **A. Deshpande**, A. Shportko, W. Fei.
Mechanistic Interpretability at ICML 2026

Talks & Posters

- Aug. 2025 [P] “Approximating Tensor Contractions with Annealed Importance Sampling.” *QSim 2025*, New York, NY.
- Dec. 2024 [P] “Quantum Circuit Volume for Graph Models.” *Illinois Mathematics Lab Open House*, Urbana, IL.

Service & Affiliations

- May 2025– Membership Director, SIAM @ University of Illinois, Urbana-Champaign
- May 2025– Member, Society for Industrial & Applied Mathematics

Coursework

Statistical Mechanics, Quantum Information Theory^G, Electromagnetism I, Classical Mechanics I & II, Special Relativity, Real Analysis, High-Dimensional Statistics^{AG}, Optimal Control Theory^{AG}, Representation Theory^{AG}, Stochastic Processes, Deep Learning Theory^G, Theoretical Neuroscience^G, Biologically-Plausible AI^G, Machine Learning, Deep Generative Models, Data Structures & Algorithms.

^G graduate coursework ^A audited

Skills

Languages: Python, C/C++, Java, Julia, Mathematica.

Libraries: NumPy, SciPy, Pandas, Matplotlib, PyTorch, SymPy.

Tools: Git, L^AT_EX, Conda, Shell, Jupyter.